PlotStudio

Minimum Stay Thresholds & Booking Potential

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QUESTION

At what minimum stay length do cancellations or low bookings spike across regions?

Analysis Preparation — Features Engineered

The dataset is now analysis-ready for minimum-stay threshold work: all `279,712` listings were preserved in a reusable feature-engineered table `df_airbnb_features`, and the unit of analysis was confirmed to be one row per listing because `listing_id` is unique across all rows.

Observations

Minimum-stay policies are heavily concentrated at very short stays, with one notable long-stay cluster at 30 nights. Globally, `1`-night minimums account for `33.19%` of listings, `2` nights for `21.76%`, `3` nights for `13.54%`, `4–6` nights for `10.76%`, and `30` nights alone for `10.33%`. That means the market is dominated by highly flexible policies, but there is also a clear monthly-stay niche that we should treat as a distinct threshold candidate later.

Extreme minimum-stay outliers exist, but they are rare enough that a 60-night main analysis window is defensible. Only `1.03%` of listings exceed `60` nights, and the empirical `99th`-percentile cap is `80` nights with just `1.00%` above it. This means we can focus most visual and threshold analysis on `minimum_nights <= 60` without losing meaningful market coverage, while still keeping capped proxy variants for sensitivity checks.

The booking-potential curve falls mechanically and sharply in the practical range. The global visual **Figure 1**

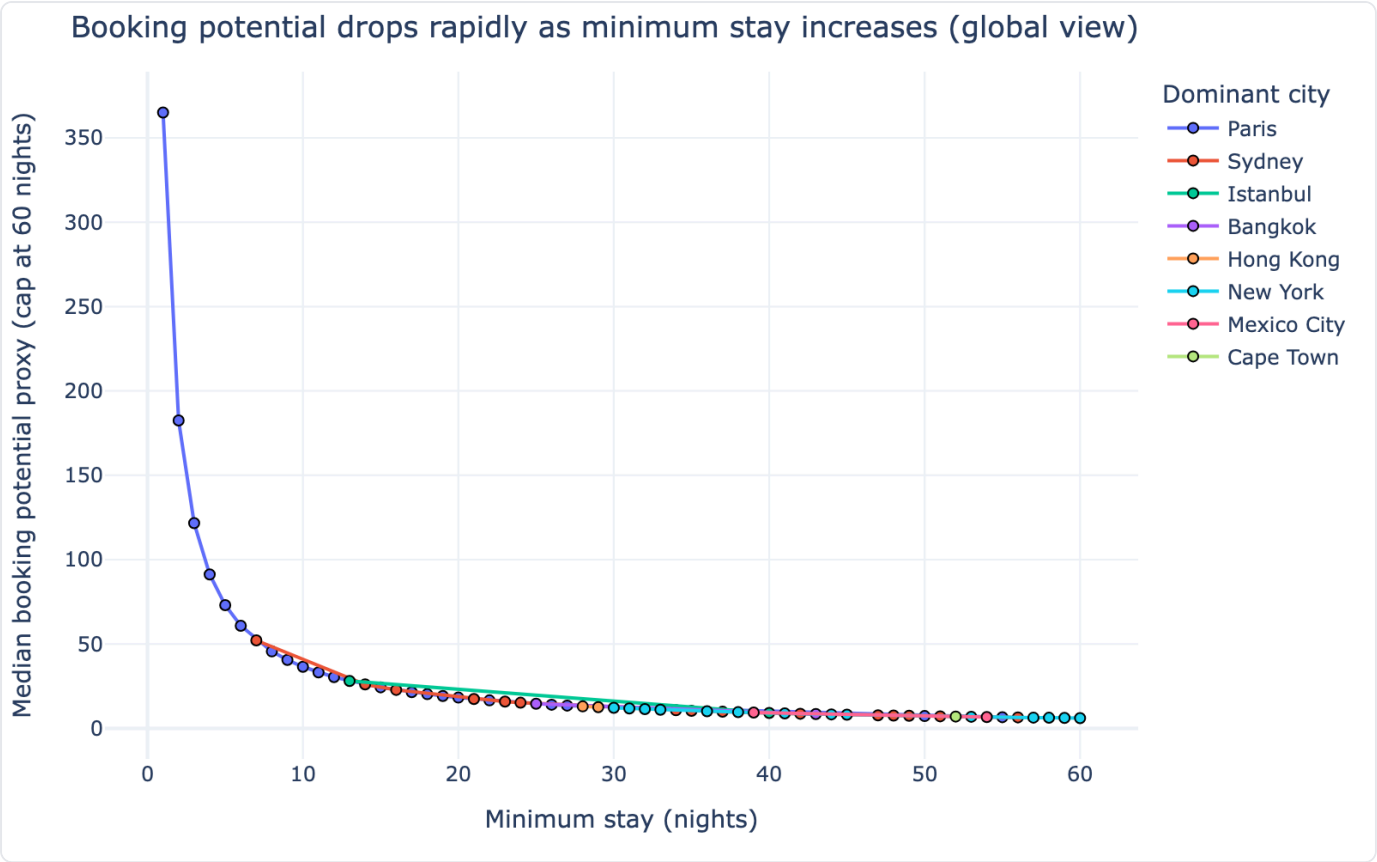

Figure 1

makes the core pattern immediately visible: booking potential is highest at very low minimum stays and drops rapidly as hosts move into longer minimum-stay settings, especially in the early part of the curve where most listings are concentrated.

Regional sample sizes are strong enough for city-level comparisons. The largest markets are Paris (`64,690` listings), New York (`37,012`), Sydney (`33,630`), and Rome (`27,647`), and every city still retains substantial coverage within the `<=60` night window—for example Paris has `64,015` listings within range and Hong Kong still has `6,987`. That gives us enough support for city-specific threshold estimates.

Most room-type segments are well represented, with only one clear caution flag. The only city-by-room-type cell under the low-count warning threshold is Rio de Janeiro hotel rooms with `99` listings. So later threshold comparisons by city and room type should generally be stable, but that one subgroup should be interpreted cautiously.

Next Steps

The next task should quantify the actual tipping points in this curve—especially around `2`, `3`, `7`, `14`, and `30` nights—and measure how large the booking-potential drop is when listings cross those thresholds globally before we break the same comparisons out by city, room type, and host segment.

Global Minimum-Stay Thresholds

Overview

Low-booking risk, using the booking-potential proxy, spikes most clearly once hosts move beyond very short stays and then again at weekly, biweekly, and monthly policy tiers. Globally, the relationship is not a single abrupt cliff; it is a steep monotonic decline with especially decision-relevant breakpoints at `2`, `3`, `7`, `14`, and `30` nights because those values combine large drops in theoretical booking flexibility with meaningful listing concentration.

The global curve in **Figure 2**

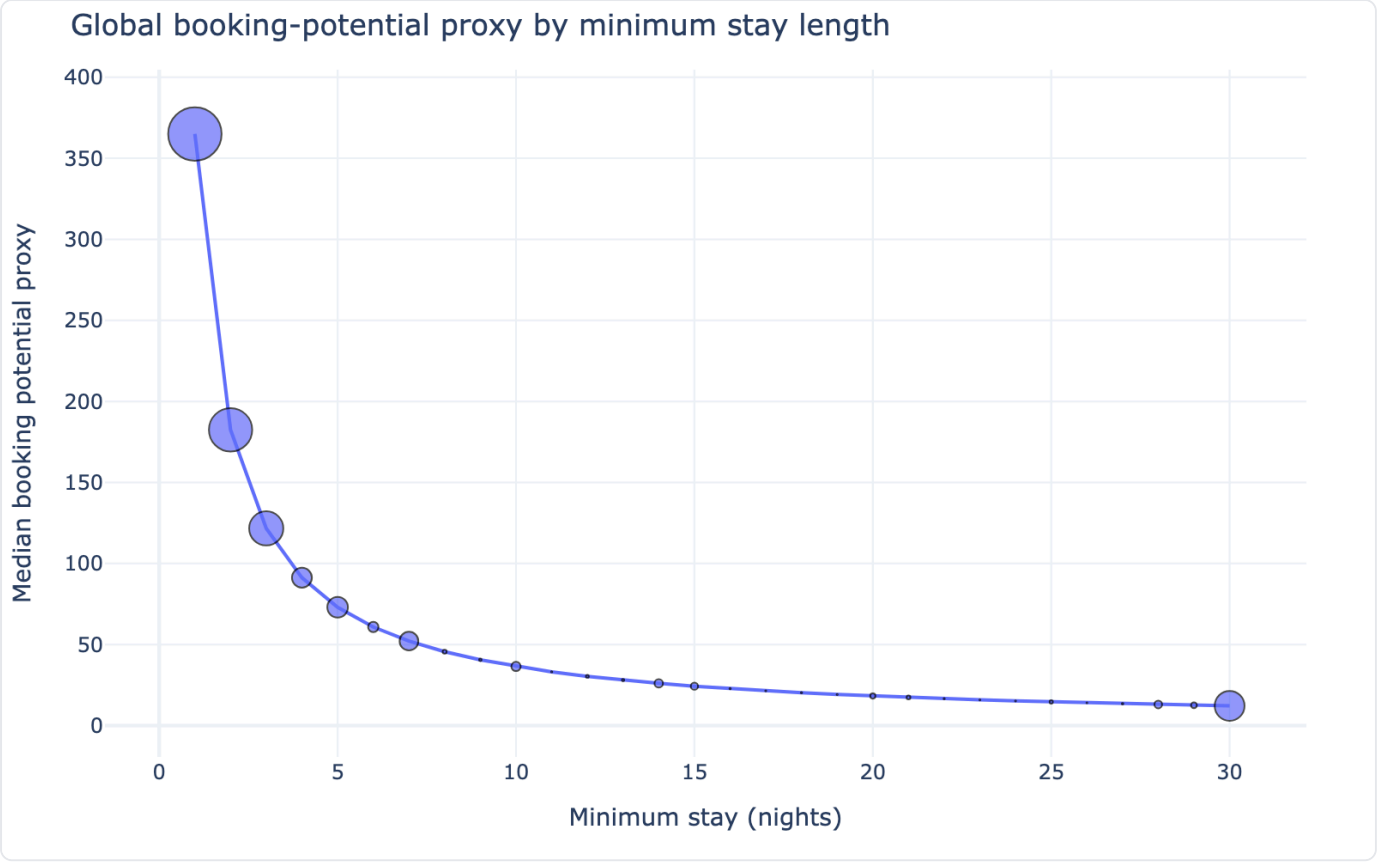

Figure 2

makes that pattern visible, while the exact-value summary `df_min_nights_summary` and threshold effect table `df_threshold_effects` quantify how severe those drops are. In practice, this means a host who raises minimum stay from the ultra-flexible range into week-long or month-long policies is accepting a major reduction in potential stay turnover, which is a strong proxy for lower booking opportunity unless they are deliberately targeting longer-stay demand.

Key Patterns & Observations

The biggest single global drop is from `1` to `2` nights. That step cuts the median booking-potential proxy from `365.00` to `182.50`, a `50.00%` decline.

Transition	Median Booking Potential Before	After	Absolute Drop	Percent Drop
1 -> 2 nights	365.00	182.50	182.50	50.00%
2 -> 3 nights	182.50	121.67	60.83	33.33%
3 -> 4 nights	121.67	91.25	30.42	25.00%
4 -> 5 nights	91.25	73.00	18.25	20.00%
6 -> 7 nights	60.83	52.14	8.69	14.29%
7 -> 14 nights	52.14	26.07	26.07	50.00%
14 -> 30 nights	26.07	12.17	13.90	53.31%

The practical threshold zone is really `1–7` nights, not just one number. The early part of the curve is where booking flexibility collapses fastest: `1-3` night policies have a median-of-medians booking potential of `182.50`, versus `73.00` for `4-6` nights and `34.84` for `7-14` nights.

Policy Band	Listings	Share of <=60-Night Listings	Median of Medians
1-3 nights	191,585	69.21%	182.50
4-6 nights	30,097	10.87%	73.00
7-14 nights	17,713	6.40%	34.84
15-29 nights	6,786	2.45%	16.59
30-60 nights	30,646	11.07%	8.20

That matters because nearly `7 in 10` listings are still in the `1-3` night band, so moving out of that zone is not a marginal tweak; it is a shift away from the dominant short-stay market.

Weekly, biweekly, and monthly policies are the clearest operational thresholds. The exact-value table shows heavy clustering at `7` nights (`11,356` listings, median proxy `52.14`), `14` nights (`2,330`, `26.07`), and especially `30` nights (`28,881`, `12.17`). Those are not just arbitrary numbers on a smooth curve; they are common host policy choices that align with sharp cumulative reductions in booking opportunity.

Common Policy Value	Listings	Median Booking Potential
7 nights	11,356	52.14
14 nights	2,330	26.07
30 nights	28,881	12.17

Broad threshold comparisons confirm that week-plus and two-week-plus minimums are where lost booking potential becomes especially large in practical terms.

Comparison	Left Group N	Right Group N	Left Median	Right Median	Median Difference	Median % Difference
1 night vs 2+ nights	92,839	183,988	365.00	121.67	243.33	200.00%
2-3 nights vs 4-7 nights	98,746	41,453	182.50	73.00	109.50	150.00%
<=7 nights vs >7 nights	233,038	43,789	182.50	12.17	170.33	1400.00%
<=14 nights vs >14 nights	239,395	37,432	182.50	12.17	170.33	1400.00%
7-14 nights vs 15-30 nights	17,713	35,667	52.14	12.17	39.98	328.57%

Interpretation & Implications

If the goal is to maximize booking opportunity, the safest global recommendation is to stay at `3` nights or below for broad-market listings, treat `4-7` nights as a meaningful restriction, and regard `14+` nights as a long-stay niche strategy rather than a mainstream setting. The evidence suggests three practical policy regimes:

Regime	Minimum Stay Range	What It Means
Flexible short-stay	1-3 nights	Highest booking opportunity; best aligned with the bulk of the market
Moderately restrictive	4-7 nights	Major reduction in flexibility, but still usable for hosts balancing turnover and cleaning costs
Long-stay oriented	14+ nights	Strongly reduced booking opportunity; better seen as a deliberate niche strategy

This framing matters for the original question because we do not observe actual cancellations or bookings here. Instead, we are identifying where minimum-stay rules begin to mimic low-utilization risk by sharply shrinking the theoretical number of stays a listing can accept. On that basis, the strongest global spike points are: `2` nights relative to `1`, `4` nights relative to `3`, the move into `7+` nights, the move into `14+` nights, and the especially restrictive `30`-night tier.

The bottom line is that **minimum-stay policies above `7` nights materially constrain booking opportunity in almost every practical sense, and `14` to `30` nights should be treated as specialized long-stay positioning rather than default host policy.**

Regional And Segment Threshold Variation

Overview

The global threshold pattern is remarkably consistent across regions and segments: booking potential falls sharply once listings move beyond `2-3` nights, falls again around `8-14` nights, and collapses in the `15-30` night band. What varies by market is not the direction of the effect, but how common those restrictive policies are and whether long-minimum listings appear premium enough to justify the lost flexibility.

The city threshold summary `df_city_thresholds` and city comparison chart **Figure 3**

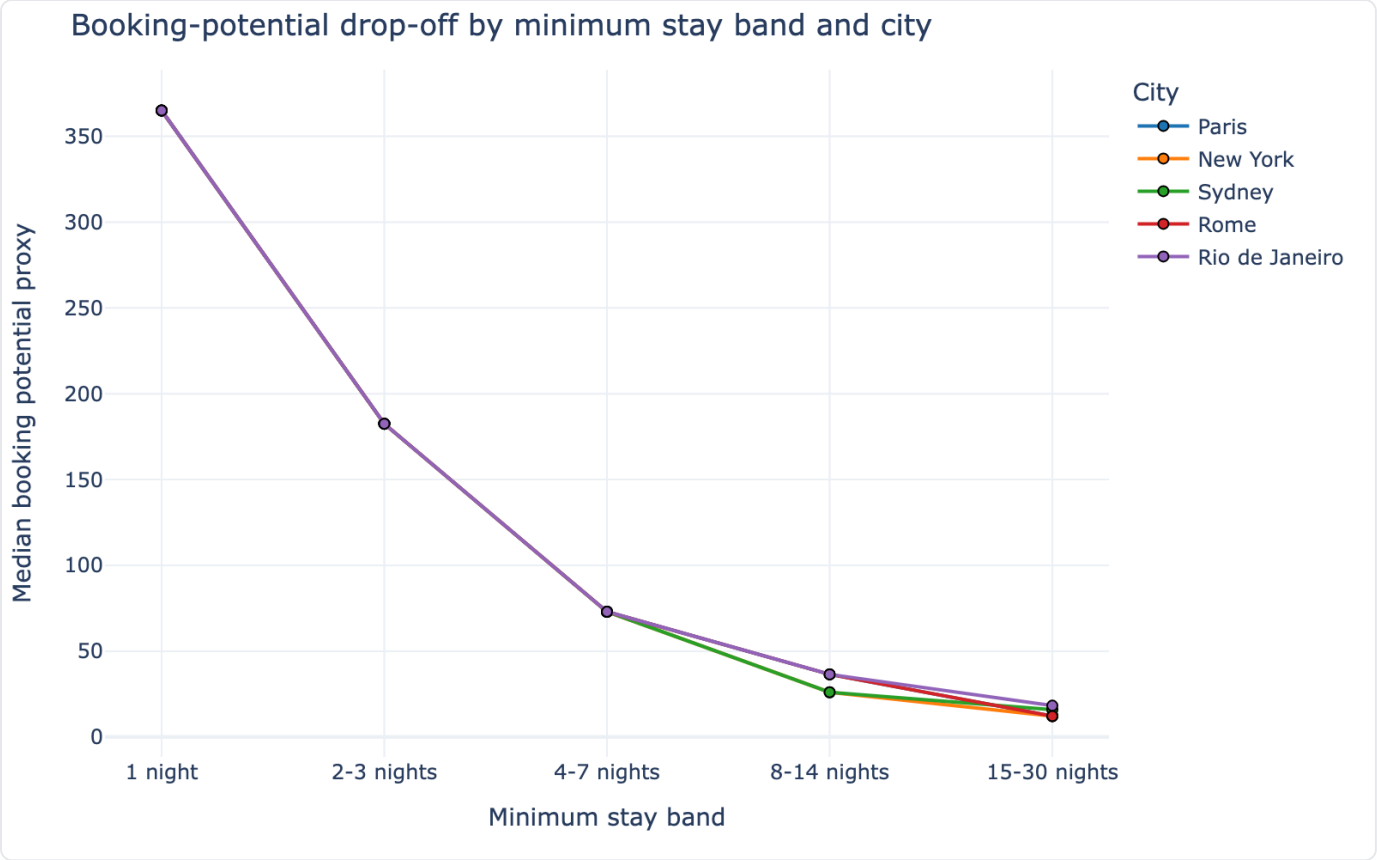

Figure 3

show that all major cities follow the same downward staircase. The segment threshold summary `df_segment_thresholds` shows that host types and room types mostly differ in how often they choose higher minimum stays, not in escaping the underlying math of reduced booking opportunity.

Key Patterns & Observations

Major cities all show the same threshold staircase, but some are steeper in the later bands. For the five largest cities, median booking potential is `182.50` in the `2-3 nights` band, around `73.00` in `4-7 nights`, about `26.07` to `36.50` in `8-14 nights`, and only `12.17` to `18.25` in `15-30 nights`.

City	2-3 Nights Median	4-7 Nights Median	8-14 Nights Median	15-30 Nights Median
New York	182.50	73.00	26.07	12.17
Paris	182.50	73.00	36.50	12.17
Rio de Janeiro	182.50	73.00	36.50	18.25
Rome	182.50	73.00	36.50	12.17
Sydney	182.50	73.00	26.07	15.87

The steepest early-band city drops are concentrated outside the very largest markets. From `2-3 nights` to `4-7 nights`, the biggest median declines are in Bangkok and Hong Kong (`66.67%`), followed by Cape Town, Istanbul, and Mexico City (`60.00%`).

City	Drop: 2-3 to 4-7 Nights
Bangkok	66.67%
Hong Kong	66.67%
Cape Town	60.00%
Istanbul	60.00%
Mexico City	60.00%

The steepest later-band declines point to markets where month-long minimums are especially restrictive. From `8-14 nights` to `15-30 nights`, the largest declines are in Mexico City, Paris, and Rome (`66.67%`), with Cape Town and Istanbul just behind (`66.09%`).

City	Drop: 8-14 to 15-30 Nights
Mexico City	66.67%
Paris	66.67%
Rome	66.67%
Cape Town	66.09%
Istanbul	66.09%

Host segments experience the same decline shape, but some segments choose longer minimums more often. Non-superhost casual hosts have the highest concentration in higher minimum-stay bands: `14.54%` of their listings fall in `15-30` or `31-60` nights combined, versus `13.11%` for non-superhost professionals, `11.52%` for superhost professionals, and `10.30%` for superhost casual hosts.

Host Segment	Share in 15-30 and 31-60 Nights
Non-superhost casual	14.54%
Non-superhost professional	13.11%
Superhost professional	11.52%
Superhost casual	10.30%

Room types differ more in policy mix than in resilience. Hotel rooms are overwhelmingly short-stay (`86.04%` at `1 night`), while private rooms and shared rooms also skew strongly short. Entire places are the segment most willing to use mid-range and monthly minimums, but their booking-potential proxy still falls along the same curve.

Room Type	Share at 1 Night	Share at 15-30 Nights
Hotel room	86.04%	2.22%
Private room	48.45%	16.43%
Shared room	62.47%	13.97%
Entire place	23.89%	11.50%

City differences are partly explained by whether longer-stay listings look premium or instead look weaker and less established. Comparing `15-30 nights` to `2-3 nights` in four illustrative cities:

City	Median Price Difference	Median Amenity Difference	Review Coverage Difference	Median Review Score Difference
Mexico City	-179.0	-5.0	-25.02 pts	+1.0
Paris	-8.0	+2.0	-17.33 pts	0.0
Rio de Janeiro	+250.0	-4.0	-38.58 pts	0.0
Sydney	-22.0	-3.0	-31.06 pts	+1.0

That comparison is especially informative:

- In **Mexico City**, longer-minimum listings are cheaper, less amenity-rich, and much less reviewed, which helps explain why `15-30` night policies look particularly costly there.
- In **Paris**, the `15-30` night band is not much cheaper and is slightly more amenity-rich, but review coverage still drops meaningfully, suggesting a thinner long-stay niche rather than a premium mainstream segment.
- In **Sydney**, longer-minimum listings look somewhat less premium and much less reviewed, again consistent with narrower demand.

- In **Rio de Janeiro**, `15-30` night listings are much more expensive but also less amenity-rich and far less reviewed, which suggests a mixed long-stay niche: more upscale pricing, but not necessarily broader booking appeal.

Interpretation & Implications

The actionable takeaway is that most cities should treat `2-3 nights` as the practical mainstream target, `4-7 nights` as a meaningful restriction, and `15-30 nights` as a specialized long-stay strategy. The regional differences matter, but they do not overturn the main answer: longer minimum stays almost always reduce booking opportunity sharply.

A practical policy framing is:

Segment	Recommended Default	Caution Threshold	Long-Stay Niche Threshold
Most cities, broad-market listings	2-3 nights	4-7 nights	15-30 nights
Entire places	Up to 3 nights, occasionally 5-7	8-14 nights	15-30 nights
Private/shared rooms	1-3 nights	4-7 nights	15+ nights
Non-superhost casual hosts	1-3 nights strongly preferred	4-7 nights risky	15+ nights usually too restrictive unless intentionally targeting long stays

So the direct answer to the user's question is: **spikes in low booking potential show up globally as soon as listings move beyond `2-3` nights, become clearly pronounced by `7` nights, and intensify further at `14` and especially `30` nights.** City and segment differences mostly change how often hosts choose those thresholds and whether long-stay listings look premium enough to partially justify them, but they rarely eliminate the drop itself.

Data Sources

- df_airbnb_features
- df_min_nights_summary
- df_threshold_effects
- df_city_thresholds
- df_segment_thresholds